B › *High-fidelity PCR for sequence validation*

☐ R0194 5 × High-fidelity PCR buffer, 40 mL

(1.) Prepare 10 µL PCR master mix for each sample:

Ingredient	Stock	Final	For 1 sample	For 8 samples
High-fidelity PCR buffer	5 ×	1 ×	2.0 µL	16.0 µL
Betaine *(optional)*	5 M	500 mM	1.0 µL	8.0 µL
$MgCl_2$	50 mM	2.5 mM	0.5 µL	4.0 µL
Forward primer	10 µM	500 nM	0.5 µL	2.0 µL
Reverse primer	10 µM	500 nM	0.5 µL	2.0 µL
dNTP mix	10 mM	1.0 mM	1.0 µL	8.0 µL
DNA polymerase, diluted	0.04 U/µL	0.008 U/µL	0.2 µL	1.6 µL
Water, reagent-grade			4.0 µL	32.0 µL
Template	5 ng/µL	1 ng/µL	0.2 µL	1.6 µL

Critical: Avoid exceeding 2 ng of a 5 kbp plasmid per reaction. Excess template can inhibit the polymerase and promote nonspecific amplification. High-fidelity enzymes are effective with as little as 1 pg/µL DNA.

(2.) *Critical:* Include a reaction without the template as negative control in each PCR run.

(3.) Spin down the reactions briefly and place in the thermocycler. Run the following program:

Cycles	Temperature	Duration	Name	Remarks
1 ×	97 °C	1:00 min	Initial denaturation	Set to 2:00 min for GC-rich templates
25–30 ×	97 °C	0:30 min	Denaturation	
	65–68 °C	0:30 min	Annealing	Set to the primer melting temperature
	72 °C	0:15 min/kbp	Extension	Adjust for target length
1 ×	72 °C	5:00 min	Final extension	
1 ×	12 °C	∞	Hold	

Critical: Consult the manufacturer's recommended cycling conditions.

(4.) Proceed with analysis of PCR products as described below.

(5.) If the amplicon will be used for restriction-ligation cloning SOP 0033, purify over a silica column SOP 0021 or by gel extraction SOP 0022 to remove primers, dNTPs, and polymerase before proceeding with restriction digest.

Recipe (available online) Troubleshooting (available online) Notes (available online)